

Occurrence-based text representations and topic modeling

Prof. Luca Cagliero
Dipartimento di Automatica e Informatica
Politecnico di Torino



**Politecnico
di Torino**

Lecture goal

- Occurrence-based text representation
- Topic modeling
 - Latent Semantic Indexing
 - Latent Dirichlet Allocation
 - The Author-Topic Model

Structured text representation

- Feature-based text representation
- Encode the information hidden in the text in a feature-value representation
- Two main strategies
 - Occurrence-based representation
 - textual features are computed on the frequencies of occurrences of the main textual units (e.g., words, n-grams, phrases) in the input text
 - Distributed vector representations of text
 - high-dimensional text representation extracted by means of neural network training

Structured text representation

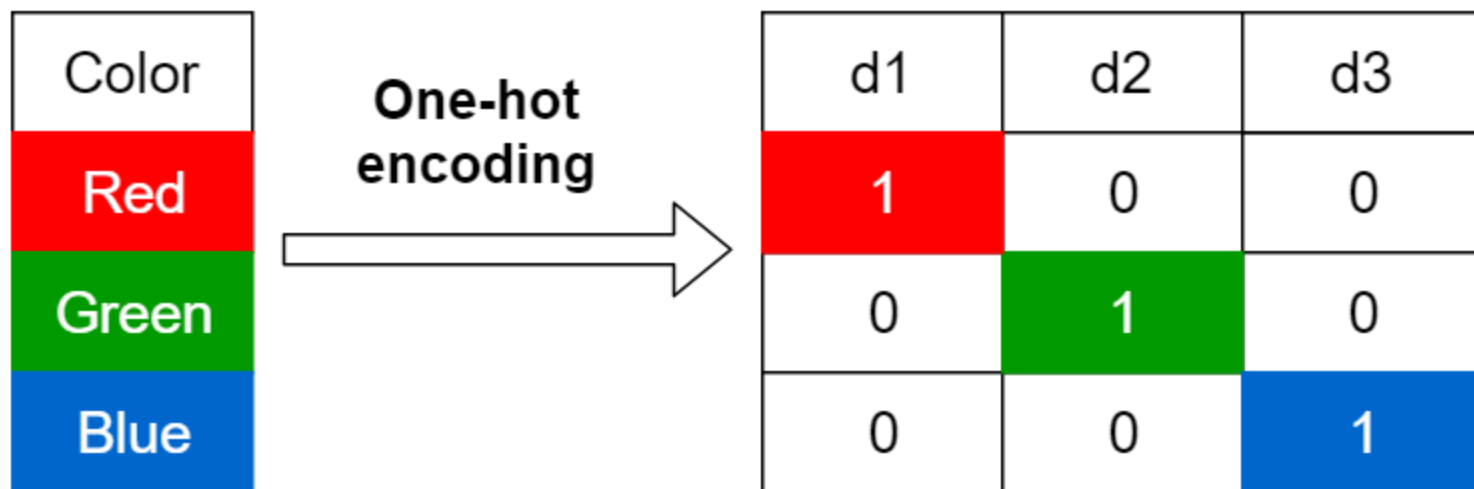
- Feature-based text representation
- Encode the information hidden in the text in a feature-value representation
- Two main strategies
 - Occurrence-based representation
 - textual features are computed on the frequencies of occurrences of the main textual units (e.g., words, n-grams, phrases) in the input text
 - Distributed vector representations of text
 - high-dimensional text representation extracted by means of neural network training

Here we focus on occurrence-based text representations

One-hot encoding

- Each word in the dictionary is regarded as discrete symbol and mapped to a distinct feature
- Each document is represented as a tuple in a relation whose schema consists of the word-level features
- The feature takes binary values (0/1) indicating whether a particular word occurs in the given document

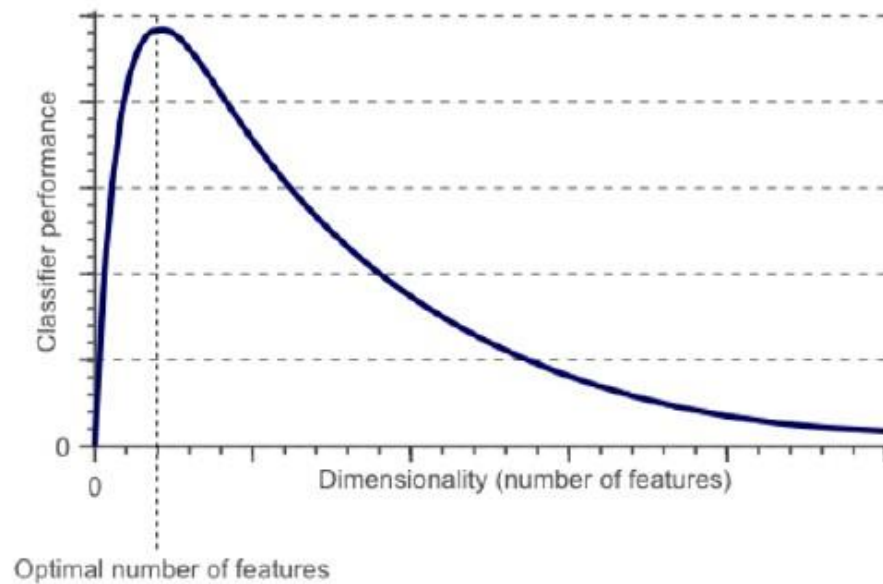
One-hot encoding



One-hot encoding: drawbacks

- High dimensionality
 - Number of words in the dictionary $\gg$ Number of documents
- Data sparsity
 - Each document is likely to include a limited number of words in the dictionary

Curse of dimensionality



Term-frequency - inverse document frequency

- Established weighted representation of text
 - Counteract data sparsity
- For each combination of word and document it stores the term frequency-inverse document frequency (tf-idf) statistics
- $n_{i,j}$: number of occurrences of word i in document d_j
- d_j : number of words in document j
- $|D|$: number of documents
- Term frequency:

$$tf_{i,j} = \frac{n_{i,j}}{d_j}$$

- Inverse document frequency:

$$idf_i = \log_{10} \frac{|D|}{|\{d:i \in d\}|}$$

- Tf-idf statistics:

$$tf_{i,j} \cdot idf_i$$

Term-frequency - inverse document frequency

- Term frequency: observed frequency of occurrence within the specific document
 - Each word is representative of a local pattern
 - Term frequency positively contributes to the relative word importance
- Document frequency: number of documents in which the word occurs at least once
 - Word occurrences are spread over all documents
 - Document frequent negatively contributes to the relative word importance

Suitable for analyzing heterogeneous document collections

Term frequency - document frequency

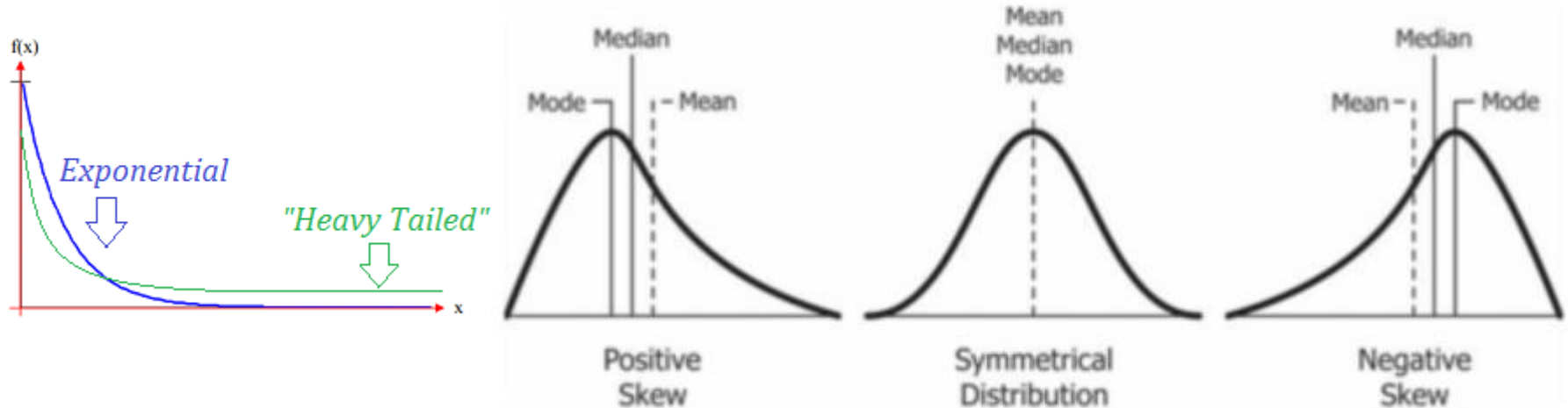
- Replace the inverse document frequency with the document frequency in the tf-idf statistics
- Word occurrences spread over all the documents are rewarded

Suitable for analyzing homogeneous document collections

Elena Baralis, Luca Cagliero, Alessandro Fiori, and Paolo Garza. 2015. MWI-Sum: A Multilingual Summarizer Based on Frequent Weighted Itemsets. ACM Trans. Inf. Syst. 34, 1, Article 5 (October 2015), 35 pages. DOI: <https://doi.org/10.1145/2809786>

Best Matching 25

- Overcome the issues related to unbounded term frequency values
 - Heavy tailed or skewed distributions of tf-idf values may come out
- It defines a family of scoring functions
 - Each of them contains slightly different components and parameters



Best Matching 25

- Key idea: introduce corpus-level statistics in the tf-idf formulation
 - *Avgdl*: average document length (expressed as number of words) over all the documents in the collection
 - k_1 : term frequency saturation (free parameter)
 - b : penalty score associated with the document length (free parameter)

$$\text{BM25}_{i,j} = \text{idf}_i \cdot \frac{tf_{i,j} \cdot (k_1 + 1)}{tf_{i,j} + k_1 \cdot (1 - b + b \cdot \frac{d_j}{\text{avgdl}})}$$

Alternative scoring functions: term frequency weight

weighting scheme	tf weight
binary	0, 1
raw count	$f_{t,d}$
term frequency	$f_{t,d} / \sum_{t' \in d} f_{t',d}$
log normalization	$\log(1 + f_{t,d})$
double normalization 0.5	$0.5 + 0.5 \cdot \frac{f_{t,d}}{\max_{\{t' \in d\}} f_{t',d}}$
double normalization K	$K + (1 - K) \frac{f_{t,d}}{\max_{\{t' \in d\}} f_{t',d}}$

Alternative scoring functions: document frequency weight

weighting scheme	idf weight ($n_t = \{d \in D : t \in d\} $)
unary	1
inverse document frequency	$\log \frac{N}{n_t} = -\log \frac{n_t}{N}$
inverse document frequency smooth	$\log \left(\frac{N}{1 + n_t} \right) + 1$
inverse document frequency max	$\log \left(\frac{\max_{\{t' \in d\}} n_{t'}}{1 + n_t} \right)$
probabilistic inverse document frequency	$\log \frac{N - n_t}{n_t}$

Alternative scoring functions

- Log normalization
 - To scale the skewed weight distributions
- Frequency max
 - To bound the range of variation of the frequency values
- Double normalization
 - To set a unique free parameter K

Additional reading on term weighting



- Christopher D. Manning, Prabhakar Raghavan and Hinrich Schütze, Introduction to Information Retrieval, Cambridge University Press. 2008.
 - <https://nlp.stanford.edu/IR-book/information-retrieval-book.html>
- Download and read chapter 6
 - <https://nlp.stanford.edu/IR-book/pdf/06vect.pdf>

Limitations of occurrence-based text representations

- Suitable for evaluating syntactical text similarities
- Unsuitable for capturing semantic text similarities

king = [0000000000010000]

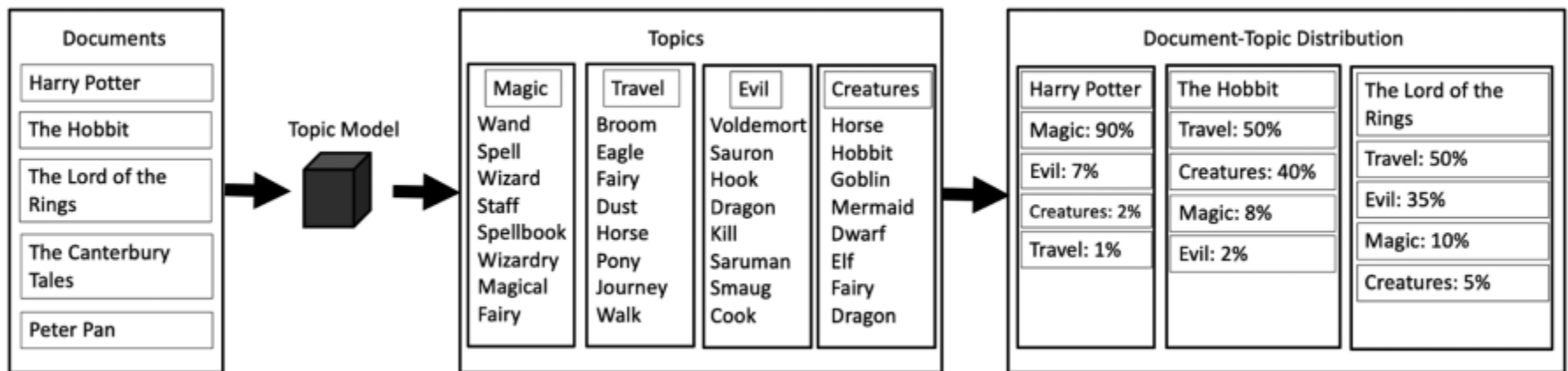
queen = [0000000010000000]

Curse of dimensionality: countermeasures

- Feature selection
 - Discard the less relevant features
- Dimensionality reduction
 - Derive a lower-dimensional representation

Topic modelling

- Established NLP application
 - Mainly unsupervised
- Focus on capturing latent topics in a large document corpus
- Provide end-users with a topic-level description of the analyzed documents



Topic modelling and dimensionality reduction

- The number of topics covered by a document is likely small
- Some embedding techniques and dimensionality reduction methods inherently provide a synthetic description of the most salient document topics
 - Vector representations of text allow clustering text into homogenous groups
 - Occurrence-based representations are high-dimensional can be reduced to low-dimensional spaces

Topic modeling techniques based on embedding techniques
will be covered later on in this course

Topic modelling: traditional approaches

- Based on occurrence-based text representation
- The distribution of words in each document is expressed as a weighted combination of concepts derived from the occurrence-based text representation
- Latent Semantic Indexing (LSI)
 - The underlying concepts and the corresponding weights are derived from the Singular Value Decomposition
- Latent Dirichlet Allocation (LDA)
 - Derive document-topics and topic-terms probabilities distributions using a generative process based on statistical Bayesian inference

Latent Semantic Indexing

- Mathematical approach used to represent occurrence-based text representations in a lower dimensional latent space
- Based on Singular Value Decomposition
 - Matrix factorization

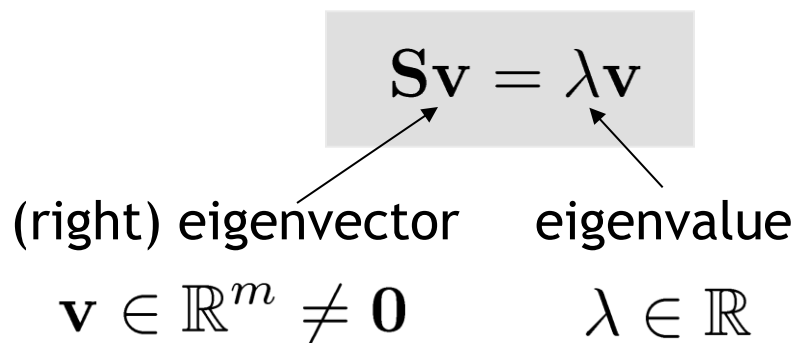
Singular Value Decomposition: preliminaries

- Eigenvectors and eigenvalues of $M \times N$ matrix A

$$\mathbf{S}\mathbf{v} = \lambda\mathbf{v}$$

(right) eigenvector eigenvalue

$\mathbf{v} \in \mathbb{R}^m \neq \mathbf{0}$ $\lambda \in \mathbb{R}$



- $\mathbf{S}$ is a square matrix
- Is this the case of a document-word matrix? Not actually..

Singular Value Decomposition: preliminaries

$$\mathbf{S}\mathbf{v} = \lambda\mathbf{v} \iff (\mathbf{S} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$$

only has a non-zero solution if $|\mathbf{S} - \lambda\mathbf{I}| = 0$

Finding the m distinct solutions (i.e., the roots of the characteristic polynomial) can be complex even though $\mathbf{S}$ is real.

Singular Value Decomposition

- For an $M \times N$ matrix $\mathbf{A}$ of rank r there exists a factorization as follows:

$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$$

The diagram illustrates the dimensions of the matrices in the SVD equation $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$. Below the equation, three boxes are arranged horizontally. The first box contains $M \times M$ and has an arrow pointing to $\mathbf{U}$. The second box contains $M \times N$ and has an arrow pointing to $\mathbf{\Sigma}$. The third box contains $V \text{ is } N \times N$ and has an arrow pointing to $\mathbf{V}^T$.

Singular Value Decomposition

$$A = U \Sigma V^T$$

Diagram illustrating the dimensions of the matrices in the SVD equation $A = U \Sigma V^T$:

- U is $M \times M$
- Σ is $M \times N$
- V is $N \times N$

- $AA^T = (U \Sigma V^T)(U \Sigma V^T)^T = (U \Sigma V^T)(V \Sigma U^T) = U \Sigma^2 U^T$
- The columns of U are orthogonal eigenvectors of AA^T
- The columns of V are orthogonal eigenvectors of $A^T A$

Singular Value Decomposition

$$A = U \Sigma V^T$$

Diagram illustrating the dimensions of the matrices in the SVD equation $A = U \Sigma V^T$:

- U is $M \times M$
- Σ is $M \times N$
- V is $N \times N$

Eigenvalues $\lambda_1 \dots \lambda_r$ of AA^T are the eigenvalues of $A^T A$

$$\sigma_i = \sqrt{\lambda_i}$$

$$\Sigma = \text{diag}(\sigma_1 \dots \sigma_r)$$

$$\sigma_{i+1} \leq \sigma_i$$

Christopher D. Manning and Pandu Nayak. Introduction to Information Retrieval. CS276

Low-rank approximation

- We exploit SVD to compute the optimal low-rank approximation
 - A lower dimensional representation of the occurrence-based text representation
- Approximation problem: find A_k of rank k such that it minimizes the Frobenius norm

$$A_k = \min_{X: \text{rank}(X)=k} \|A - X\|_F$$

$$\|A\|_F \equiv \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2}.$$

- A_k and X are $M \times N$ matrices
- The rank $r \gg K$ reflects the underlying dimensionality of the data

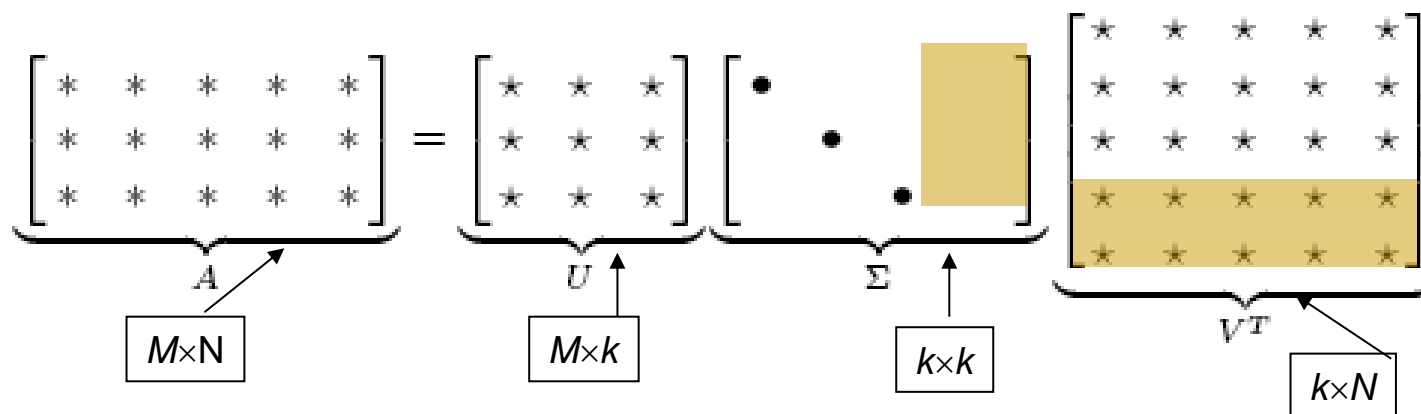
Low-rank approximation

- The approximation, in terms of the Frobenius norm of the error, is

$$\min_{X: \text{rank}(X)=k} \|A - X\|_F = \|A - A_k\|_F = \sigma_{k+1}$$

Low-rank approximation

- We retain only the top- k singular values
 - All the other are reduced to zero
- We get a **reduced SVD**

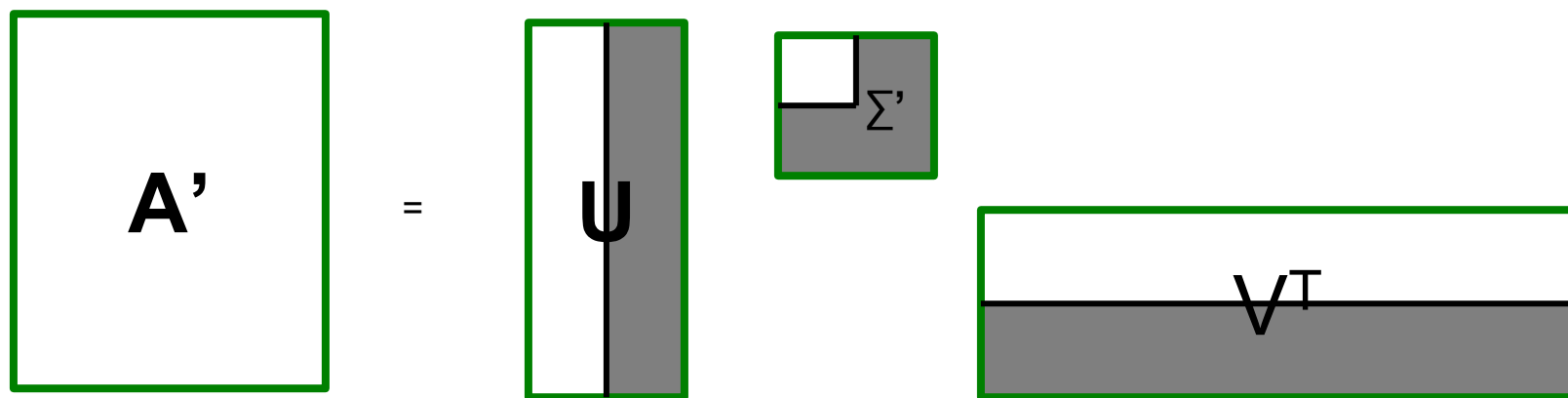
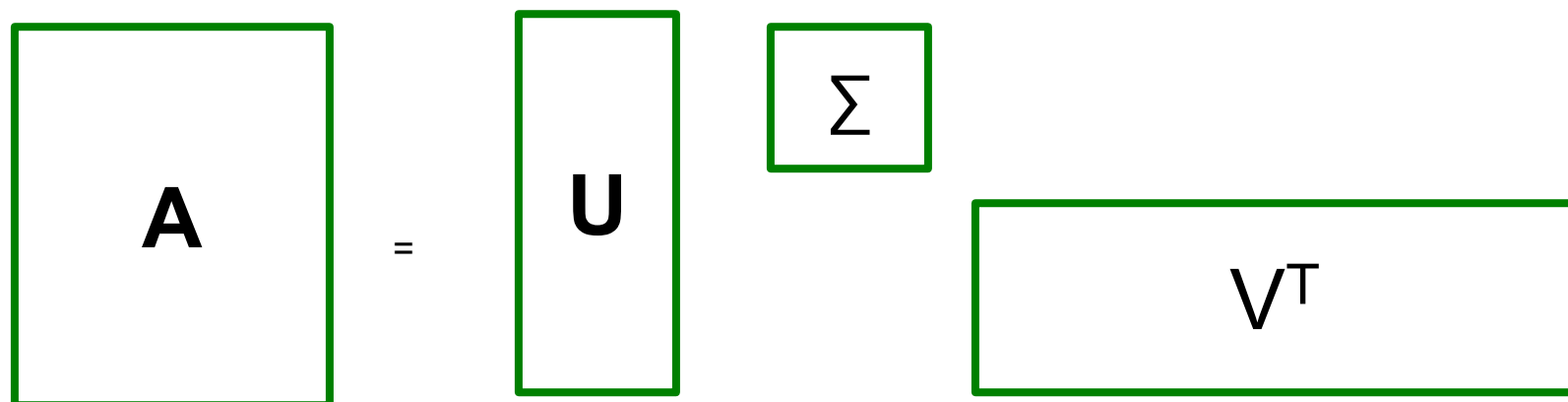


The diagram illustrates the reduced SVD decomposition of a matrix A . On the left, matrix A is shown as a 3x5 grid of asterisks, with a label $M \times N$ below it. This is followed by an equals sign. To the right of the equals sign are three matrices: U , Σ , and V^T . Matrix U is a 3x3 grid of asterisks, labeled $M \times k$ below it. Matrix Σ is a 3x3 grid containing three dots on the diagonal and a yellow shaded rectangle in the bottom-right corner, labeled $k \times k$ below it. Matrix V^T is a 3x5 grid of asterisks, with the bottom two rows highlighted in yellow, labeled $k \times N$ below it. Arrows point from the labels A , U , Σ , and V^T to their respective matrices.

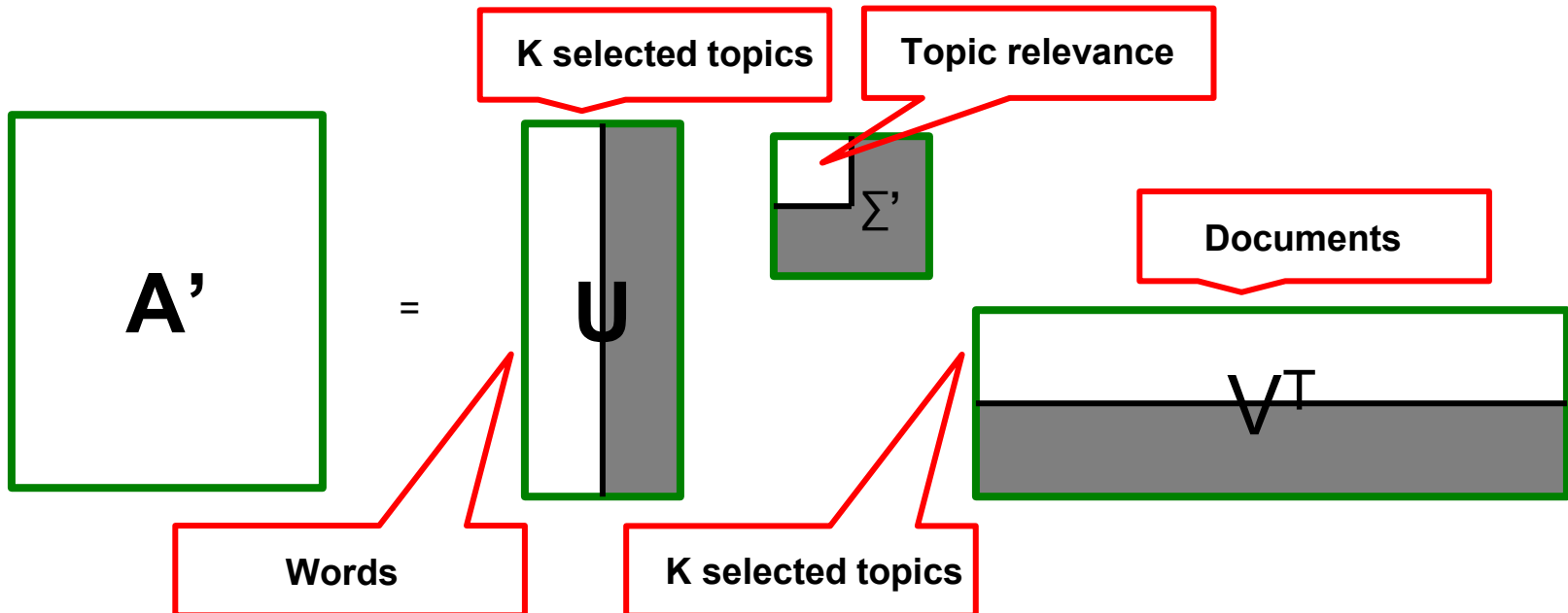
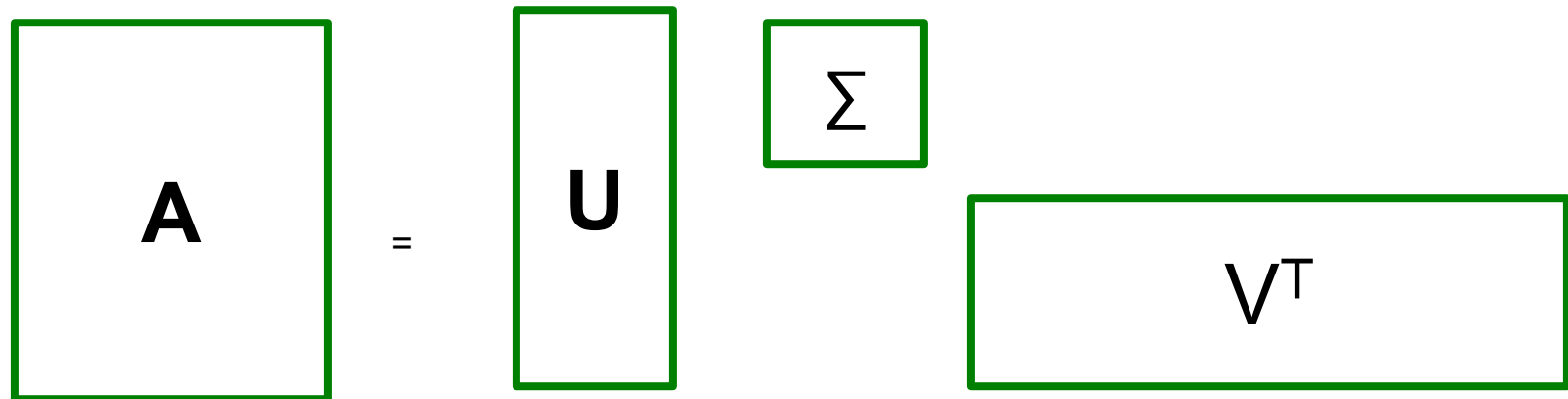
$$\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} \bullet & & \\ & \bullet & \\ & & \bullet \end{bmatrix} \begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix}$$

$M \times N$ $M \times k$ $k \times k$ $k \times N$

Singular Value Decomposition



Singular Value Decomposition



Latent Semantic Indexing via SVD

- Compute a low-rank approximation of the document-word matrix
- Reduce the data dimensionality
 - Maps terms and documents to a lower dimensional representation
 - Preserve pairwise word associations conveyed by the occurrence-based model

$$\min_{X: \text{rank}(X)=k} \|A - X\|_F = \|A - A_k\|_F = \sigma_{k+1}$$

Singular Value Decomposition

- Document similarity is computed based on the inner product in the low-dimensional representation
- The cosine similarity between the original document vectors can be computed as $A'^T A' = V' \Sigma^T \Sigma' V'^T$

Topic modeling using LSA

- LSA is instrumental for topic detection
- Each eigenvalue in A' is representative of a salient document concept
 - The i -th concept corresponding to σ_i is likely to be more relevant than those associated with σ_{i+1}

Topic modeling using LSA

- The first k rows of V^T represent the relevance of each document to the shortlisted top- k concepts
 - We multiply the diagonal matrix Σ' by a specific orthogonal eigenvector in V^T
- The first k columns of U represent the relevance of each word to the shortlisted top- k concepts
 - We multiply the specific orthogonal eigenvector in U by the diagonal matrix Σ'

Topic modeling using LSA: example

C	d_1	d_2	d_3	d_4	d_5	d_6
ship	1	0	1	0	0	0
boat	0	1	0	0	0	0
ocean	1	1	0	0	0	0
wood	1	0	0	1	1	0
tree	0	0	0	1	0	1

Topic detection based on LSI: example

U	1	2	3	4	5	
ship	−0.44	−0.30	0.00	0.00	0.00	
boat	−0.13	−0.33	0.00	0.00	0.00	
ocean	−0.48	−0.51	0.00	0.00	0.00	
wood	−0.70	0.35	0.00	0.00	0.00	
tree	−0.26	0.65	0.00	0.00	0.00	
Σ_2	1	2	3	4	5	
1	2.16	0.00	0.00	0.00	0.00	
2	0.00	1.59	0.00	0.00	0.00	
3	0.00	0.00	0.00	0.00	0.00	
4	0.00	0.00	0.00	0.00	0.00	
5	0.00	0.00	0.00	0.00	0.00	
V^T	d_1	d_2	d_3	d_4	d_5	d_6
1	−0.75	−0.28	−0.20	−0.45	−0.33	−0.12
2	−0.29	−0.53	−0.19	0.63	0.22	0.41
3	0.00	0.00	0.00	0.00	0.00	0.00
4	0.00	0.00	0.00	0.00	0.00	0.00
5	0.00	0.00	0.00	0.00	0.00	0.00

How can we set up the number of topics?

- The value K for truncated SVD is analyst-provided
- Heuristic approach
 - define σ_k as the smallest singular value that is above the half of the highest one (i.e., $\sigma_k > \sigma_1/2$)

Josef Steinberger and Karel Ježek. 2005. Text Summarization and Singular Value Decomposition. Springer, Berlin, Berlin, 245–254. DOI:http://dx.doi.org/10.1007/978-3-540-30198-1_25

SVD complexity

- $O(\min(NM^2, MN^2))$
- The complexity is lower when
 - the matrix is sparse
 - we consider only the $k \ll M$ singular vectors (reduced SVD)
 - Typical situation while coping with textual data
- Efficient SVD implementation are available, e.g.,
 - MATLAB (<https://www.mathworks.com/help/>)
 - SK-Learn (<https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.TruncatedSVD.html>)
 - Hadoop Spark (<https://spark.apache.org/docs/2.2.0/mllib-dimensionality-reduction.html>)
 - R (<https://www.rdocumentation.org/packages/base/versions/3.6.2/topics/svd>)

LSA – Exercise 1

Let us consider the following SVD decomposition of a word-document matrix R (i.e., rows are words $w_1, w_2, \dots, w_6$ whereas columns are documents $d_1, d_2, \dots, d_5$).

Find the document-level topic relevance scores

$$\begin{array}{c} R \\ \begin{bmatrix} 5 & 5 & 0 & 0 & 1 \\ 4 & 5 & 1 & 1 & 0 \\ 5 & 4 & 1 & 1 & 0 \\ 0 & 0 & 4 & 4 & 4 \\ 0 & 0 & 5 & 5 & 5 \\ 1 & 1 & 4 & 4 & 4 \end{bmatrix} \end{array} \approx \begin{array}{c} U \\ \begin{bmatrix} -0,27 & 0,55 & -0,78 & 0 \\ -0,29 & 0,47 & 0,44 & -0,71 \\ -0,29 & 0,47 & 0,44 & 0,71 \\ -0,45 & -0,29 & -0,01 & 0 \\ -0,56 & -0,36 & -0,02 & 0 \\ -0,50 & -0,18 & -0,05 & 0 \end{bmatrix} \end{array} \begin{array}{c} \Sigma \\ \begin{bmatrix} 13,74 & 0 & 0 & 0 \\ 0 & 10,88 & 0 & 0 \\ 0 & 0 & 1,36 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array} \begin{array}{c} V^T \\ \begin{bmatrix} -0,32 & -0,32 & -0,52 & -0,52 & -0,5 \\ 0,63 & 0,63 & -0,25 & -0,25 & -0,29 \\ -0,02 & -0,02 & 0,41 & 0,41 & -0,82 \\ 0,71 & -0,71 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Solution

$$\begin{matrix} & R & & U & & \Sigma & & C_1 & C_2 & & V^T \\
 \begin{bmatrix} 5 & 5 & 0 & 0 & 1 \\ 4 & 5 & 1 & 1 & 0 \\ 5 & 4 & 1 & 1 & 0 \\ 0 & 0 & 4 & 4 & 4 \\ 0 & 0 & 5 & 5 & 5 \\ 1 & 1 & 4 & 4 & 4 \end{bmatrix} & \approx & \begin{bmatrix} -0,27 & 0,55 & -0,78 & 0 \\ -0,29 & 0,47 & 0,44 & -0,71 \\ -0,29 & 0,47 & 0,44 & 0,71 \\ -0,45 & -0,29 & -0,01 & 0 \\ -0,56 & -0,36 & -0,02 & 0 \\ -0,50 & -0,18 & -0,05 & 0 \end{bmatrix} & & \begin{bmatrix} 13,74 & 0 & 0 & 0 \\ 0 & 10,88 & 0 & 0 \\ 0 & 0 & 1,36 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & & \begin{bmatrix} -0,32 & -0,32 \\ 0,63 & 0,63 \\ -0,02 & -0,02 \\ 0,71 & -0,71 \end{bmatrix} & & \begin{bmatrix} -0,52 & -0,52 & -0,5 \\ -0,25 & -0,25 & -0,29 \\ 0,41 & 0,41 & -0,82 \\ 0 & 0 & 0 \end{bmatrix}
 \end{matrix}$$

- How can we choose the optimal value of K?
 - Heuristic approach -> K = 2
- Example: d_1 topic relevance vector
 - Topic #1
 - $r_1 \times c_1 = -4.31$
 - Topic #2
 - $r_2 \times c_1 = 6.85$

LSA – Exercise 2

Let us consider the following SVD decomposition of a word-document matrix R (i.e., rows are words $w_1, w_2, \dots, w_6$, columns are documents $d_1, d_2, \dots, d_5$).

Rank the input documents in order of decreasing relevance to topic #2

What are the most representative words for that topic?

$$\begin{matrix} R & U & \Sigma & V^T \\ \begin{bmatrix} 5 & 5 & 0 & 0 & 1 \\ 4 & 5 & 1 & 1 & 0 \\ 5 & 4 & 1 & 1 & 0 \\ 0 & 0 & 4 & 4 & 4 \\ 0 & 0 & 5 & 5 & 5 \\ 1 & 1 & 4 & 4 & 4 \end{bmatrix} & \approx \begin{bmatrix} -0,27 & 0,55 & -0,78 & 0 \\ -0,29 & 0,47 & 0,44 & -0,71 \\ -0,29 & 0,47 & 0,44 & 0,71 \\ -0,45 & -0,29 & -0,01 & 0 \\ -0,56 & -0,36 & -0,02 & 0 \\ -0,50 & -0,18 & -0,05 & 0 \end{bmatrix} & \begin{bmatrix} 13,74 & 0 & 0 & 0 \\ 0 & 10,88 & 0 & 0 \\ 0 & 0 & 1,36 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} -0,32 & -0,32 & -0,52 & -0,52 & -0,5 \\ 0,63 & 0,63 & -0,25 & -0,25 & -0,29 \\ -0,02 & -0,02 & 0,41 & 0,41 & -0,82 \\ 0,71 & -0,71 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

LSA - Ex. 2 solution

$$\begin{matrix} & R & & & U & & \Sigma & & V^T \\ & & & r_1 & & & c_2 & & \\ \begin{bmatrix} 5 & 5 & 0 & 0 & 1 \\ 4 & 5 & 1 & 1 & 0 \\ 5 & 4 & 1 & 1 & 0 \\ 0 & 0 & 4 & 4 & 4 \\ 0 & 0 & 5 & 5 & 5 \\ 1 & 1 & 4 & 4 & 4 \end{bmatrix} & \approx & \begin{bmatrix} -0,27 & 0,55 & -0,78 & 0 \\ -0,29 & 0,47 & 0,44 & -0,71 \\ -0,29 & 0,47 & 0,44 & 0,71 \\ -0,45 & -0,29 & -0,01 & 0 \\ -0,56 & -0,36 & -0,02 & 0 \\ -0,50 & -0,18 & -0,05 & 0 \end{bmatrix} & \begin{bmatrix} 13,74 & 0 & 0 & 0 \\ 0 & 10,88 & 0 & 0 \\ 0 & 0 & 1,36 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} -0,32 & -0,32 & -0,52 & -0,52 & -0,5 \\ 0,63 & 0,63 & -0,25 & -0,25 & -0,29 \\ -0,02 & -0,02 & 0,41 & 0,41 & -0,82 \\ 0,71 & -0,71 & 0 & 0 & 0 \end{bmatrix}
 \end{matrix}$$

- Topic relevance vectors separately for each document
 - $d_1 = [-4.31, 6.85]$
 - $d_2 = [-4.31, 6.85]$
 - $d_3 = [-7.14, -2.72]$
 - $d_4 = [-7.14, -2.72]$
 - $d_5 = [-6.87, -3.16]$
- Documents d_1 and d_2 are mostly relevant to topic #2
- Topic #2 word relevance scores separately for each word
 - $W_1: 5.984 (= r_1 \times c_2)$
 - $W_2: 5.11$
 - $W_3: 5.11$
 - $W_4: -3.16$
 - $W_5: -3.92$
 - $W_6: -1.96$

Latent Dirichlet Allocation

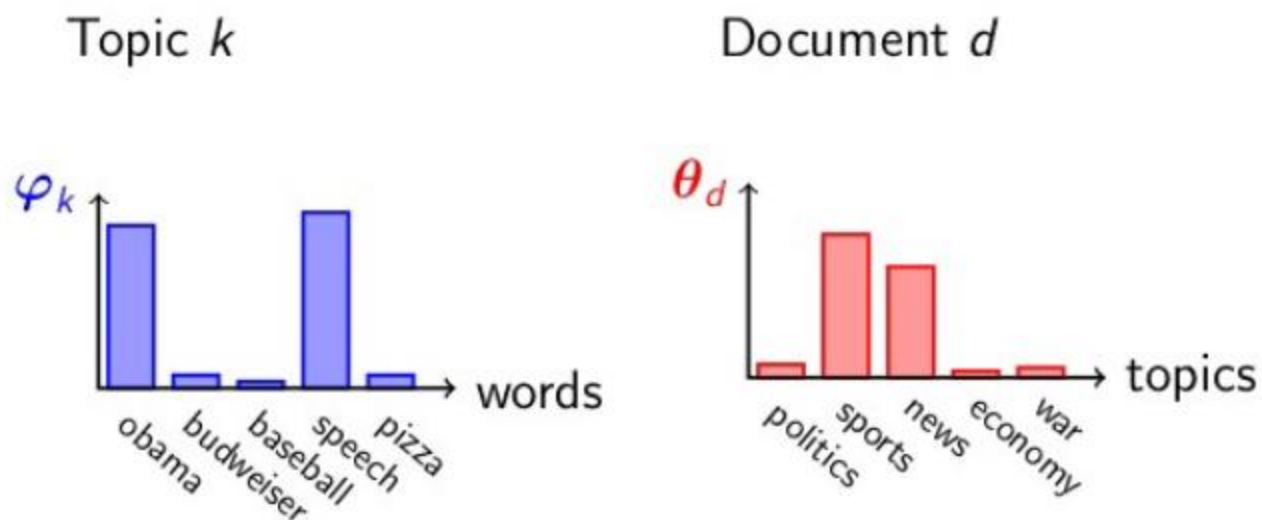
- Probabilistic topic model
- Key idea: documents are mixtures of multiple topics
- Topics are defined as distributions over a fixed vocabulary
- Documents are defined as distributions over the set of different topics
 - They range over multiple topics in different proportions

Generating Summary Keywords for Emails Using Topics. Mark Dredze, Hanna M. Wallach, Danny Puller, , Fernando Pereira. IUI'08. ACM. 2008

Latent Dirichlet Allocation

- Iterative process
- Incremental approximation of the a priori probability distribution (i.e., our prior knowledge) and the a posteriori probability distribution (the updated knowledge after exploring the data)
- The process stops at convergence
 - The n -th iteration does not produce any significant improvement compared to the $(n-1)$ -th iteration

Latent Dirichlet Allocation



David M. Blei, Andrew Y. Ng, Micheal I. Jordan. Latent Dirichlet Allocation. Journal of Machine Learning Research 3 (2003) 993-1022

Latent Dirichlet Allocation

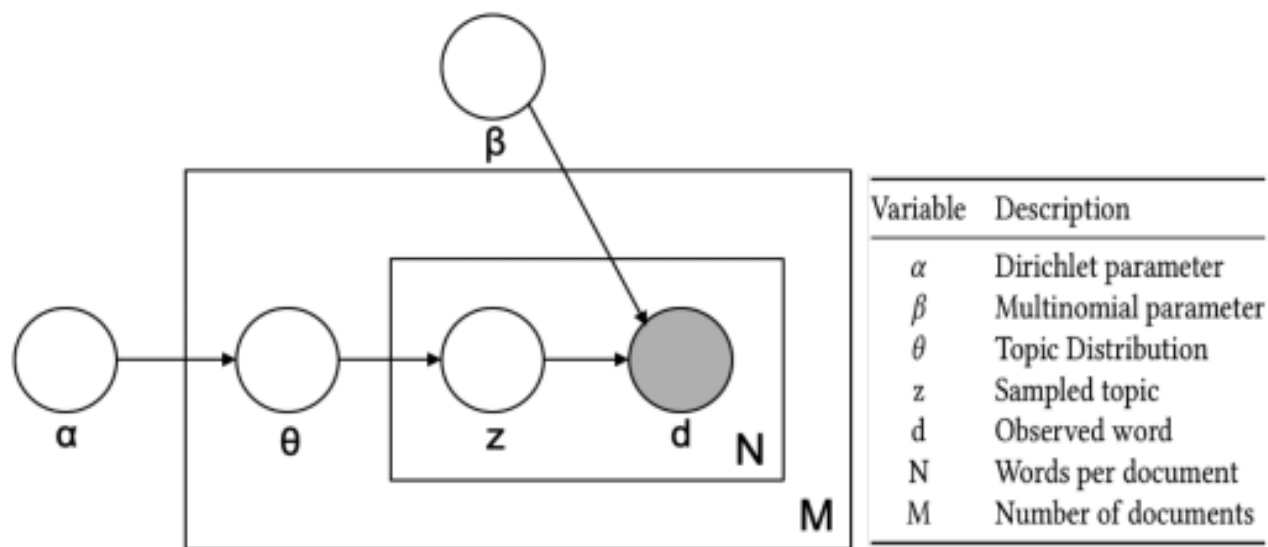
- Generative topic model
- Each word w in a document d is assumed to have been generated
 - by sampling a topic from a document-specific distribution over topics and
 - by sampling a word from the distribution over words that characterizes that topic
- The distributions over topics and words are drawn from conjugate Dirichlet priors $\text{Dir}(\alpha)$ and $\text{Dir}(\beta)$

$$g(\theta_1, \theta_2, \dots, \theta_k | (x_1, x_2, \dots, x_k)) = \text{Dir}_k(\alpha_1 + x_1, \alpha_2 + x_2, \dots, \alpha_k + x_k)$$

LDA inputs

- Vocabulary of words W
- The number of topics k
- parameters α and β
- All variables that contribute to generate the corpus are independent of the number N of words in each document

Latent Dirichlet Allocation



Circles are parameters, rectangles are iterative processes

Statistical inference

- Statistical inference algorithms are used to optimize the Dirichlet parameters and to infer the latent topics and document-specific topic mixtures
 - E.g., the Gibbs-Expectation Maximization (EM) algorithm is commonly used to do statistical inference of the posterior distribution of the latent variable for a given corpus
 - Gibbs-EM alternates between optimizing α and β and sampling a topic assignment for each word in the corpus from the distribution over topics for that word, conditioned on all other variables

Latent Dirichlet Allocation

- For each document in the corpus and for each term
 - A topic is chosen accordingly to the document-topic distribution
- Words are extracted from the input vocabulary V by considering terms probabilities for each given topic of the documents' mixture

Generative process for LDA

- Key elements
 - the corpus
 - the documents
 - the terms
- Each topic z is sampled from every word w in the document
 - the same document can be described by multiple topics
- The variables θ_d are sampled once per document
- Parameters α and β are sampled once for the whole document generating process

Generative process for LDA

- Find the parameters of a topic-word distribution that maximize the likelihood of documents in the dataset over k topics
- Each document has a topic distribution specific to it that is proportional to the probability of each of its words in the topic-word distribution
- LDA uses expectation maximization to train its model and has two main parameters aside from k : α and β
 - α corresponds to topics-per-document ratio
 - Setting α higher results in more topics per document
 - β corresponds to words-per-topic ratio
 - Setting β lower results in fewer words per topic

Generative process for LDA

For each document d in a collection D :

- (1) Randomly draw the number of words N for d .
- (2) Randomly draw the topic distribution θ from the Dirichlet distribution, conditioned on the parameter α .
- (3) For each word w_i , $0 \leq i < N$:
 - (a) Draw a topic z_i from θ .
 - (b) Draw a word w_i based on the probability of w_i given the topic z_i and conditioned on the parameter β .

Latent Dirichlet Allocation

- The probability of word w given document d and the Dirichlet priors' parameters α and β is given by

$$P(w|d, \alpha, \beta) = \sum_{t=1}^T P(w|t, \beta)P(t|d, \alpha)$$

- T is the number of latent topics

The LDA generative model

- The joint multivariate distribution of the topic mixture

$$p(\boldsymbol{\theta}, \mathbf{z}, \mathbf{w} | \boldsymbol{\alpha}, \boldsymbol{\beta}) = p(\boldsymbol{\theta} | \boldsymbol{\alpha}) \prod_{n=1}^{N_d} p(z_n | \boldsymbol{\theta}) p(w_n | z_n, \boldsymbol{\beta})$$

- The probability of the whole corpus is computed by
 - integrating over the whole distribution θ
 - summing over all topics z
 - taking the product of the marginal probabilities of each document

$$p(\mathcal{D} | \boldsymbol{\alpha}, \boldsymbol{\beta}) = \prod_{d=1}^M \int p(\boldsymbol{\theta}_d | \boldsymbol{\alpha}) \left(\prod_{n=1}^{N_d} \sum_{z_{dn}} p(z_{dn} | \boldsymbol{\theta}_d) p(w_{dn} | z_{dn}, \boldsymbol{\beta}) \right) d\boldsymbol{\theta}_d$$

LDA complexity

- High complexity on large document corpora
- Efficient implementations are available
 - MATLAB (<https://www.mathworks.com/help/textanalytics/ref/ldamodel.html>)
 - SK-Learn
(<https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.LatentDirichletAllocation.html>)
 - Hadoop Spark
(<https://spark.apache.org/docs/2.3.1/api/java/org/apache/spark/mllib/clustering/LDA.html>)

Additional reading on LDA



- David M. Blei, Andrew Y. Ng, Micheal I. Jordan. Latent Dirichlet Allocation. Journal of Machine Learning Research 3 (2003) 993-1022
- Download and read the paper:
<https://www.jmlr.org/papers/volume3/blei03a/blei03a.pdf>

The Author-Topic Model

- Generative model for documents
- It extends the Latent Dirichlet Allocation to include authorship information
- It describes the topics according to the following aspects:
 - Documents
 - Terms
 - Authors

Michal Rosen-Zvi, Thomas Griffiths, Mark Steyvers, and Padhraic Smyth. 2004. The author-topic model for authors and documents. In Proceedings of the 20th conference on Uncertainty in artificial intelligence (UAI '04). AUAI Press, Arlington, Virginia, USA, 487–494.

The Author-Topic Model

- Example of applications
 - Who is the most authoritative author on a given topic?
 - What are the topics covered by a given author?
 - What is the most authoritative paper of an author?

Michal Rosen-Zvi, Thomas Griffiths, Mark Steyvers, and Padhraic Smyth. 2004. The author-topic model for authors and documents. In Proceedings of the 20th conference on Uncertainty in artificial intelligence (UAI '04). AUAI Press, Arlington, Virginia, USA, 487–494.

Author-Topic Model

- A document d is a vector of N_d words w_d
- Each w_d is chosen from a vocabulary of size V
- A vector of A_d authors a_d is chosen from a set of authors of size A .
- A collection of D documents is defined by

$$D = \{(w_1, a_1), \dots, (w_d, a_d)\}$$

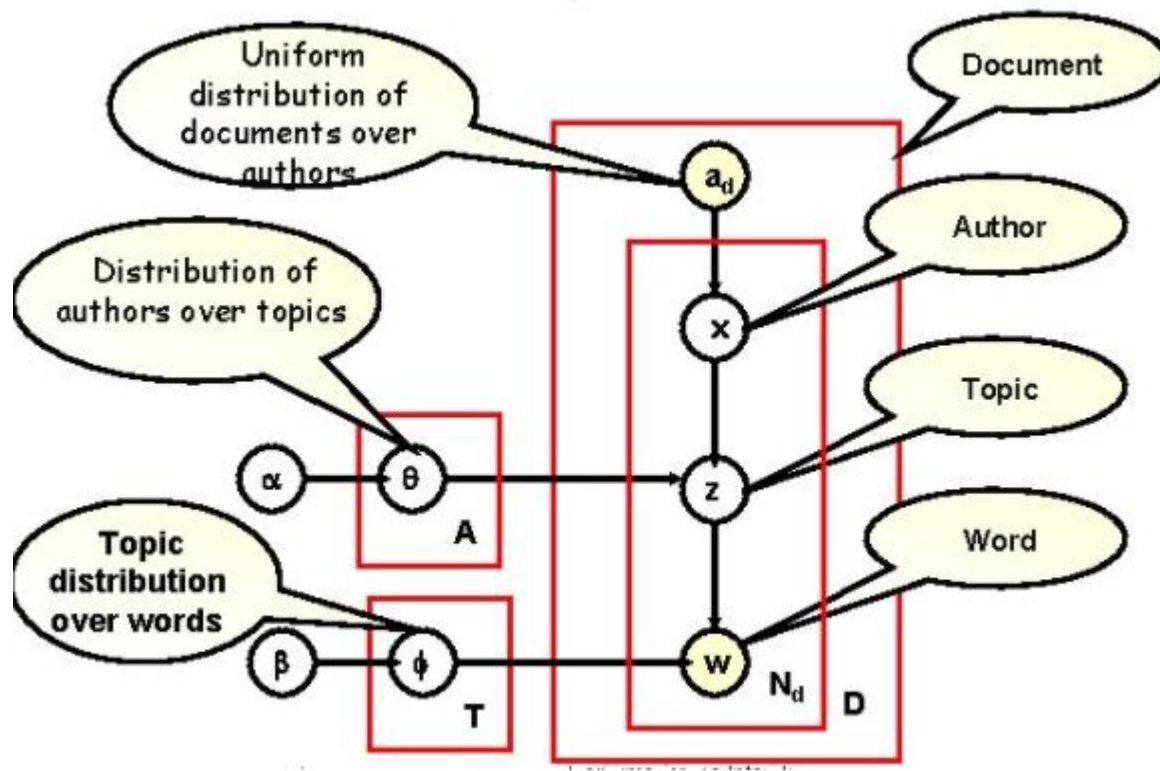
Michal Rosen-Zvi, Thomas Griffiths, Mark Steyvers, and Padhraic Smyth. 2004. The author-topic model for authors and documents. In Proceedings of the 20th conference on Uncertainty in artificial intelligence (UAI '04). AUAI Press, Arlington, Virginia, USA, 487–494.

The Author-Topic Model

- Each author is associated with a multinomial distribution over topics
- Each topic is associated with a multinomial distribution over words
- A document with multiple authors is modeled as a distribution over topics that is a mixture of the distributions associated with the authors

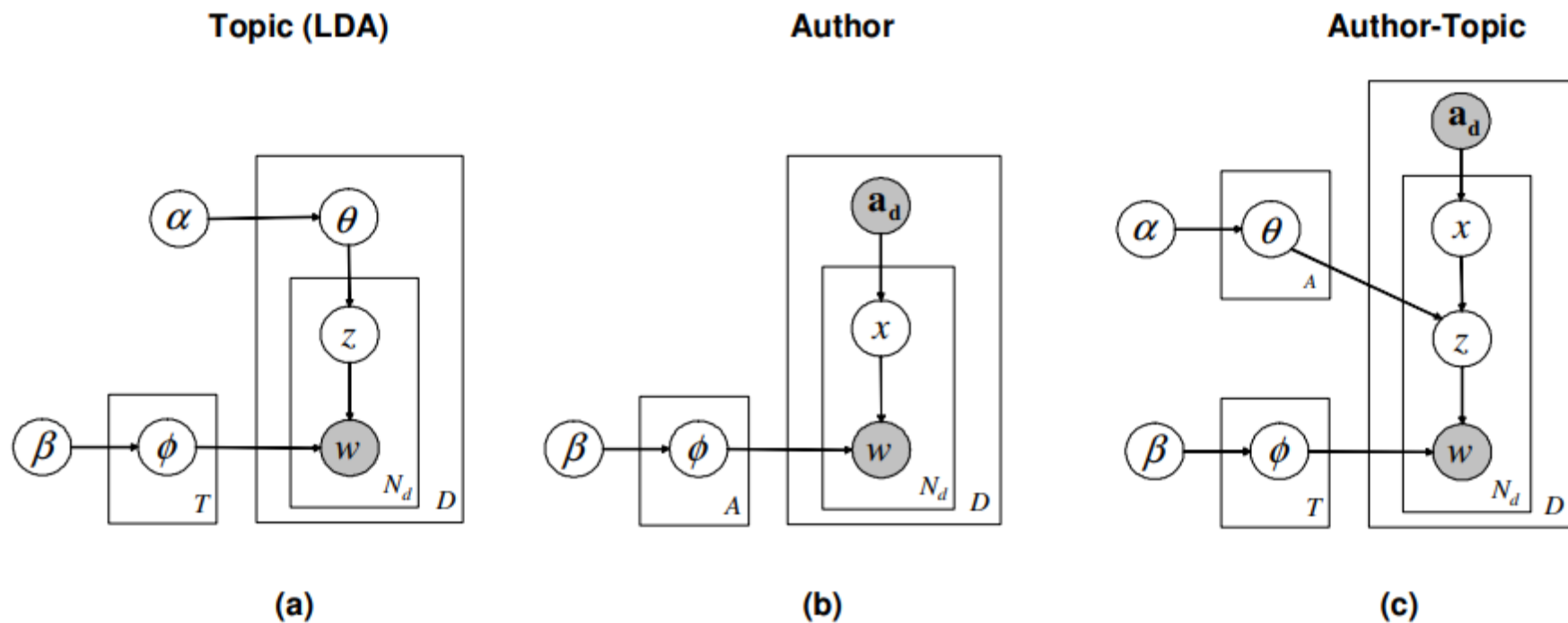
Michal Rosen-Zvi, Thomas Griffiths, Mark Steyvers, and Padhraic Smyth. 2004. The author-topic model for authors and documents. In Proceedings of the 20th conference on Uncertainty in artificial intelligence (UAI '04). AUAI Press, Arlington, Virginia, USA, 487–494.

The Author-Topic Model



Modeling Documents. Amruta Joshi. Stanford University. Department of Computer Science.

The Author-Topic Model



Michal Rosen-Zvi, Thomas Griffiths, Mark Steyvers, and Padhraic Smyth. 2004. The author-topic model for authors and documents. In Proceedings of the 20th conference on Uncertainty in artificial intelligence (UAI '04). AUAI Press, Arlington, Virginia, USA, 487–494.

Gibbs sampling for statistical inference

- Markov chain that converges to the posterior distribution on topics $\mathbf{z}$ and authors $\mathbf{x}$

$$P(z_i = j, x_i = k | w_i = m, \mathbf{z}_{-i}, \mathbf{x}_{-i}, \mathbf{w}_{-i}, \mathbf{a}_d) \propto \frac{C_{mj}^{WT} + \beta}{\sum_{m'} C_{m'j}^{WT} + V\beta} \frac{C_{kj}^{AT} + \alpha}{\sum_{j'} C_{kj'}^{AT} + T\alpha}$$

- Use the results to infer the parameters
 - Θ : probability of a given word given a topic
 - φ : probability of a topic given an author

$$\phi_{mj} = \frac{C_{mj}^{WT} + \beta}{\sum_{m'} C_{m'j}^{WT} + V\beta}$$
$$\theta_{kj} = \frac{C_{kj}^{AT} + \alpha}{\sum_{j'} C_{kj'}^{AT} + T\alpha}$$

Michal Rosen-Zvi, Thomas Griffiths, Mark Steyvers, and Padhraic Smyth. 2004. The author-topic model for authors and documents. In Proceedings of the 20th conference on Uncertainty in artificial intelligence (UAI '04). AUAI Press, Arlington, Virginia, USA, 487–494.

Additional reading on ATM



- Michal Rosen-Zvi, Thomas Griffiths, Mark Steyvers, and Padhraic Smyth. 2004. The author-topic model for authors and documents. In Proceedings of the 20th conference on Uncertainty in artificial intelligence (UAI '04). AUAI Press, Arlington, Virginia, USA, 487–494.
- Download and read the paper: <https://arxiv.org/abs/1207.4169>

Acknowledgements and copyright license



- Copyright licence
 - Attribution + Noncommercial + NoDerivatives
- Affiliation
 - The author and his staff are currently members of the Database and Data Mining Group at Dipartimento di Automatica e Informatica (Politecnico di Torino) and of the SmartData interdepartmental centre
 - <https://dbdmg.polito.it>
 - <https://smartdata.polito.it>

Thank you!